
State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

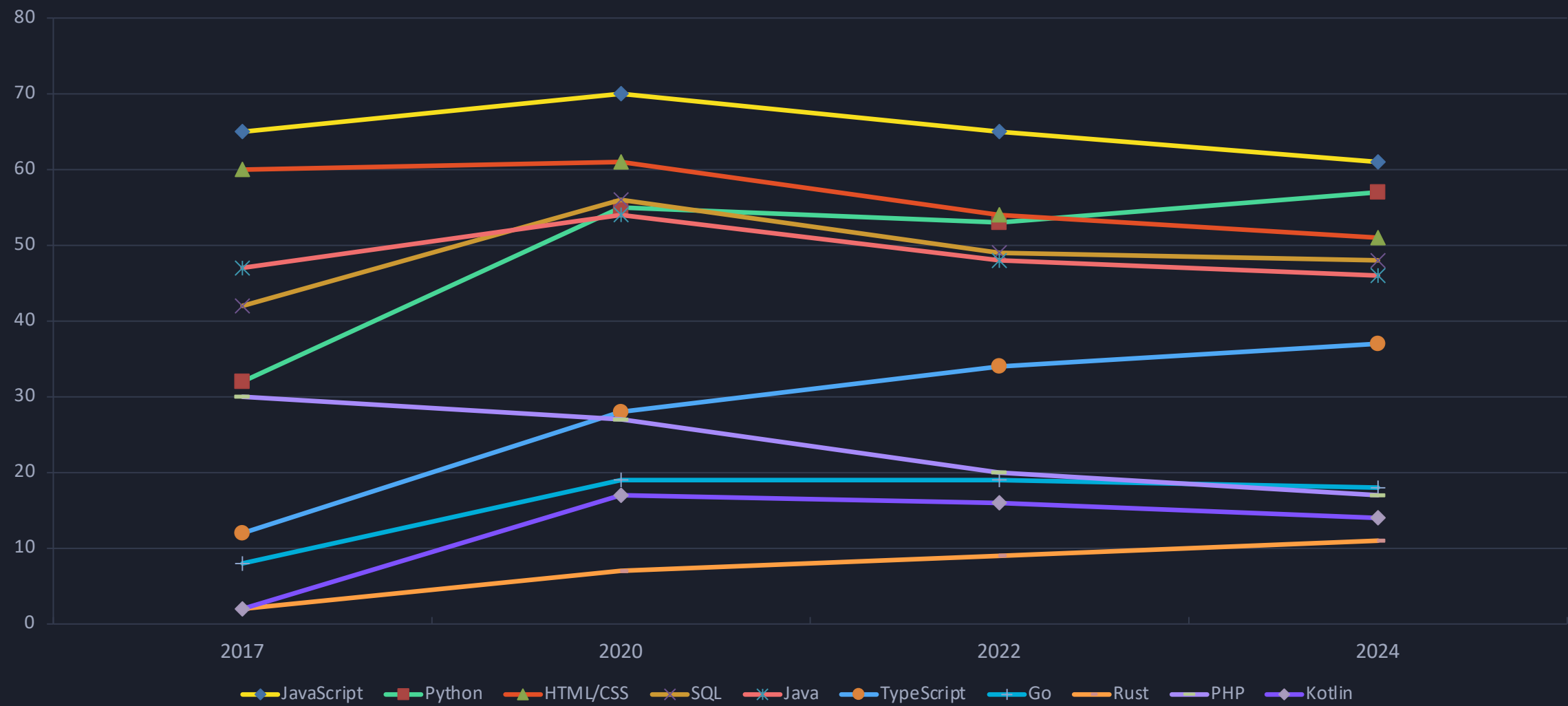
Key insights from JetBrains' survey of 23,262 developers on programming language usage, satisfaction, adoption intent, and domain specialization trends from 2017 to 2024.

Source: JetBrains State of Developer Ecosystem Report 2024

Audience: Engineering Leaders • CTOs • Developer Experience
Teams

Python Closes the Gap: 8 Years of Language Usage Shifts

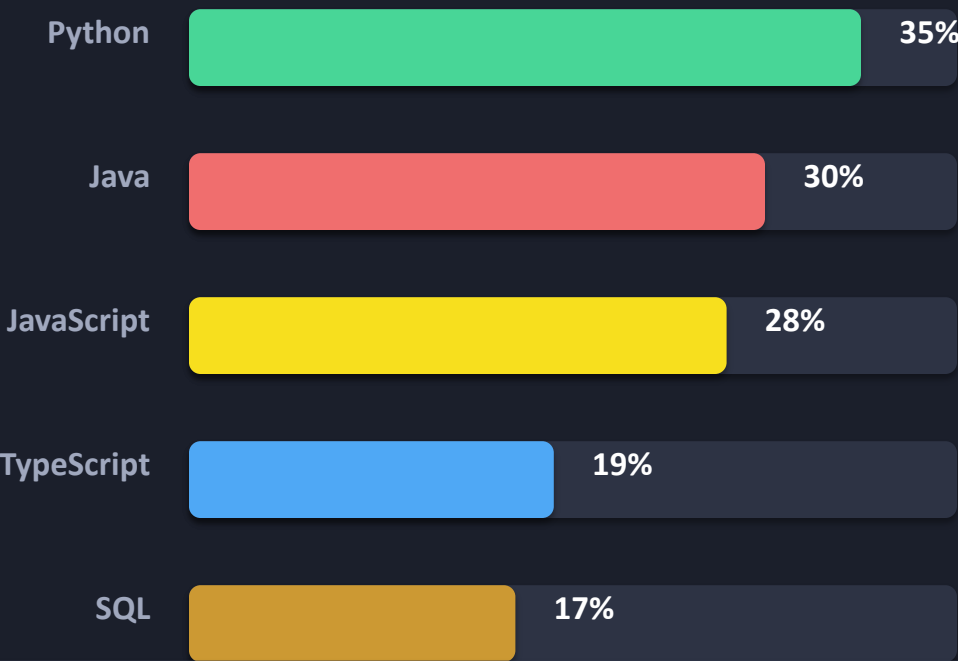
Percentage of developers using each language, 2017–2024 • n = 23,262



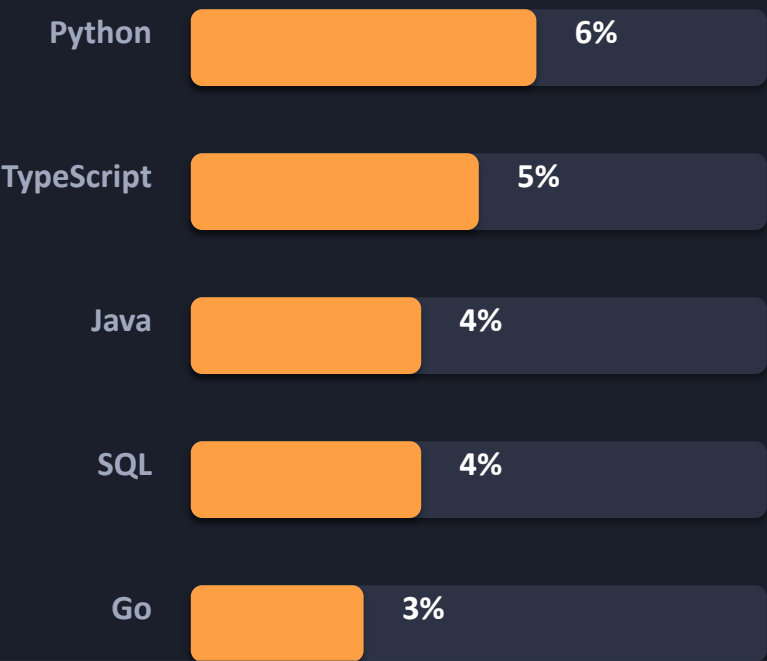
Python Is Now the #1 Primary Language at 35%

Primary language share vs. adoption intent — developers rarely switch stacks

PRIMARY LANGUAGE



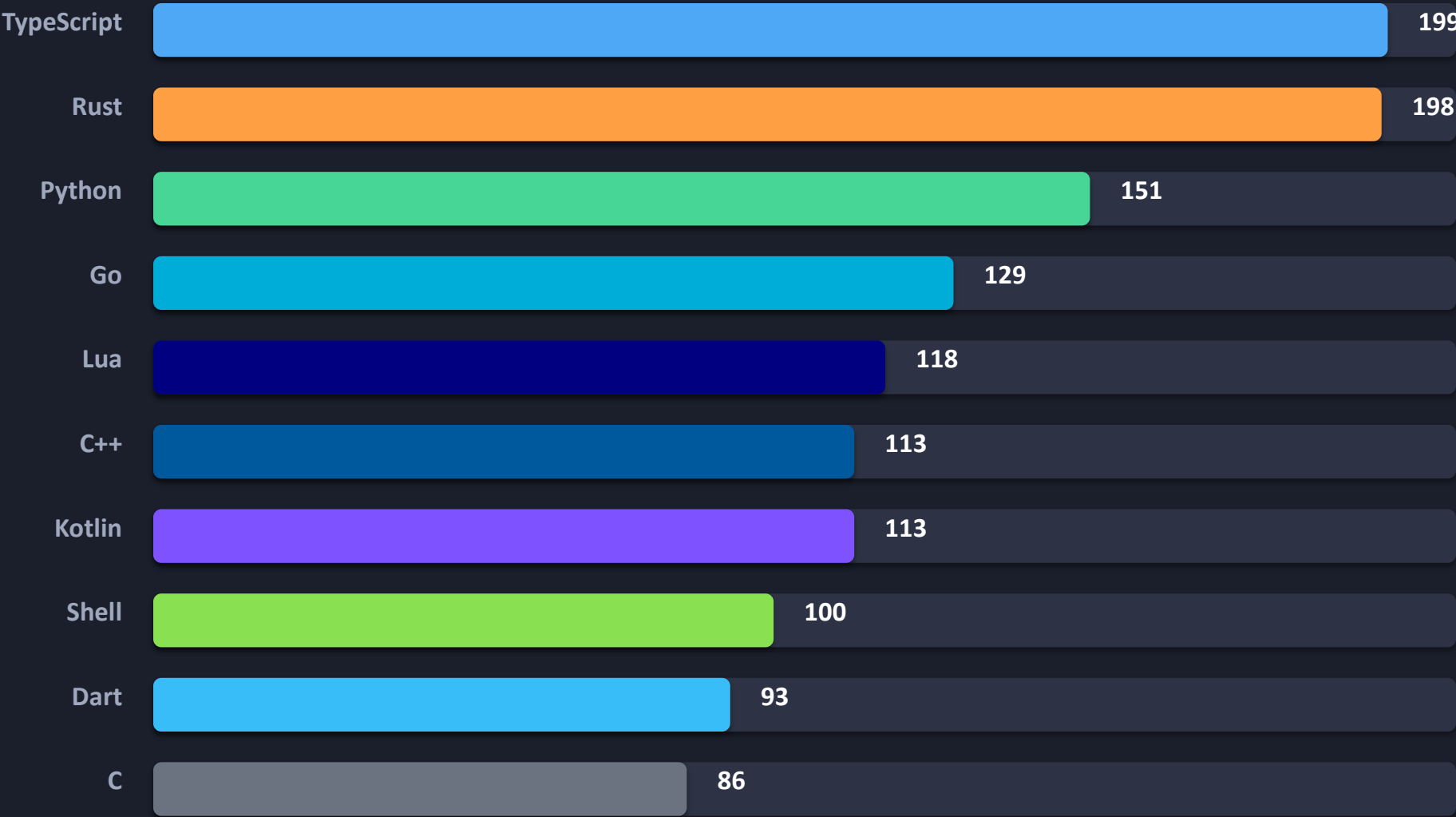
ADOPTION INTENT



Key Insight: Adoption intent remains below 6% across all languages — ecosystem lock-in and expertise investment keep developers loyal. Hiring for niche stacks will become harder over time.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Developer satisfaction scores — type-safe, modern-tooling languages dominate



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Top language by application domain — language choice is driven by domain, not general preference

DOMAIN	TOP LANGUAGE	SHARE
 AI / Machine Learning	Python	31%
 Data Science & Analytics	Python	28%
 Data Processing	Python	17%
 Integrating with APIs	Python	38%
 User Interface	TypeScript	62%
 APIs & Services	Java	49%
 Application Logic & Workflows	Java	55%
 System Tools & Components	C++	27%
 Hardware Interfacing	C++	21%
 Graphics & Rendering	C++	15%

What This Means for Your Technology Strategy in 2025

1

Invest in Python for AI/ML

Python is the undisputed leader for AI (31%), data science (28%), and data processing. Any AI/ML team investment should center on Python expertise.

2

TypeScript is the future of frontend and full-stack

Tripled from 12% to 37% usage and leads developer satisfaction at 199. It is the safest bet for UI and full-stack hiring.

3

Rust and Go are the growth plays for infrastructure

Go stabilized at 18%; Rust continues climbing to 11%. Both address cloud-native, microservices, and systems needs where performance is non-negotiable.

4

PHP and Ruby hiring pools are shrinking

PHP dropped from 30% to 17%; Ruby from 10% to 4%. Plan migration paths or accept a tightening talent market.

5

Developer satisfaction correlates with type safety

TypeScript (199) and Rust (198) top satisfaction. Modern tooling and type systems drive retention — factor this into your stack decisions.